HEALTHCARE MANAGEMENT RELATIONAL DATABASE SYSTEM

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Project Link: https://github.com/hibabtl/healthcare.db

1. Introduction

This project involved the design and implementation of a relational database using SQLite to simulate a healthcare management system. The objective was to construct a database containing at least 1000 records and incorporating multiple data types (nominal, ordinal, interval, and ratio), while demonstrating relational integrity using foreign keys and normalization principles.

Healthcare was selected as the domain because it provides a naturally structured, multi-entity environment in which relationships between patients, doctors, appointments, and prescriptions can be realistically modeled. The system was designed to reflect real-world data interactions while maintaining logical integrity, scalability, and ethical responsibility.

The final database contains over **7,000** total records across four interrelated tables.

2. Data Generation Methodology

All data was synthetically generated using Python. The following libraries were used:

- Faker (for realistic names, dates, and text data)
- Random (for controlled numerical distributions)
- Pandas
- SQLite (for database creation and constraint enforcement)

No external datasets were used or downloaded; all records were generated programmatically to ensure data originality. The final dataset includes:

- 1200 Patients
- 50 Doctors
- 3000 Appointments
- 3000 Prescriptions

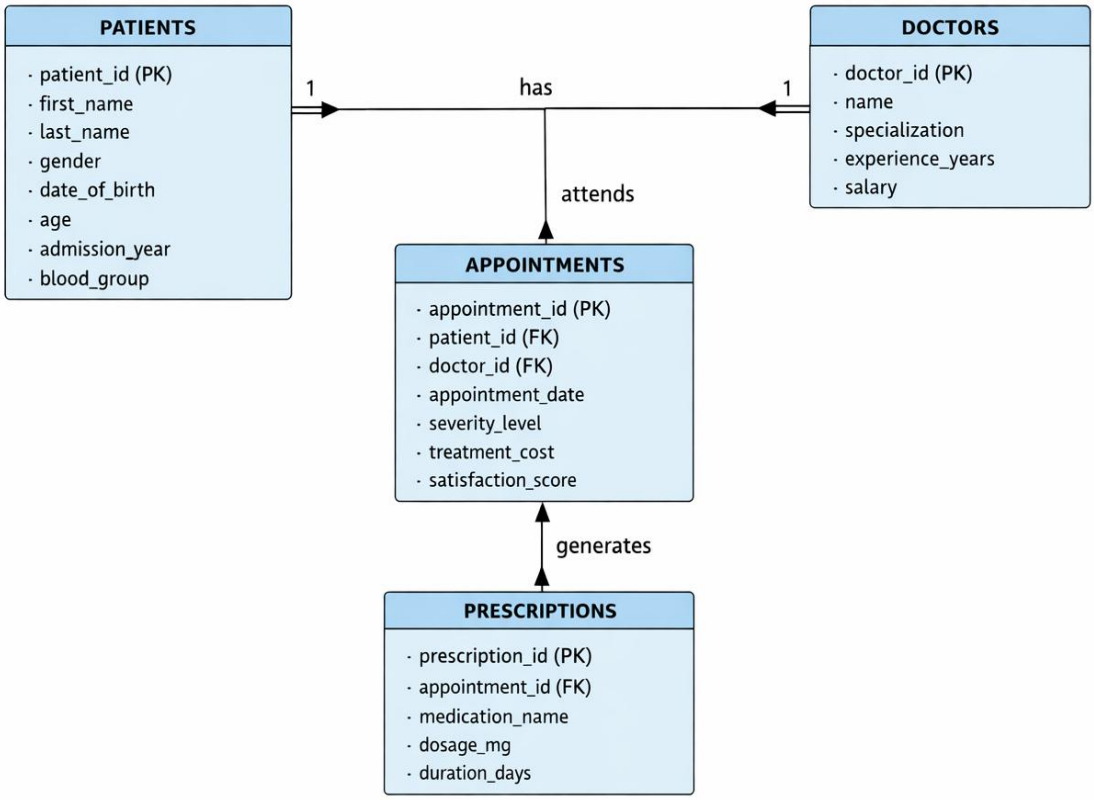
2.1 DATA TYPES

The database includes the following data types:

DATA TYPES	COLUMNS
➢ Nominal	Gender, Blood Group, Doctor Specialization
➢ Ordinal	Severity Level (Mild < Moderate < Severe < Critical)
➢ Interval	Admission Year, Satisfaction Score (1–10)
➢ Ratio	Age, Salary, Treatment Cost, Experience Years, Dosage (mg)

3. Database Schema

The database consists of four normalized tables with defined primary and foreign keys.



The ER diagram represents a normalized healthcare relational database consisting of four main entities:

- Patients
- Doctors
- Appointments
- Prescriptions

Each entity is uniquely identified by a primary key (PK), while foreign keys (FK) establish relationships between tables to maintain referential integrity.

4. Use of Multiple Tables

In this database, multiple tables are used to follow database normalization principles.

4.1 Avoid Redundancy

By separating tables:

- Data duplication is reduced
- Storage efficiency is improved
- Updates become easier

4.2 Referential Integrity

Foreign key constraints ensure:

- No appointment can exist without a valid patient
- No prescriptions can exist without a valid appointment

This maintains logical consistency across the system

4.3 Scalability

The relational design allows:

- Adding new departments
- Adding billing tables
- Expanding patients history

5. Ethical and Data Privacy Considerations

Healthcare data is among the most sensitive categories of personal information. Although this project uses synthetic data, ethical considerations remain critical.

6. Conclusion

This project successfully demonstrates the design and implementation of a relational database in SQLite that satisfies all assignment requirements. The database includes more 1000 rows, multiple related tables, appropriate data type representation (nominal, ordinal, interval, ratio), and enforced relational integrity through foreign keys. The system follows normalization principles, avoid redundancy, and reflect a realistic healthcare management system.